

ABDUL MANAN

SYSTEMS ENGINEERING / NETWORKING / SECURITY

Austin, TX · (718) 578-3399 · fnu.abdul.manan@gmail.com
linkedin.com/in/fnu-abdul-manan · abdul-manaan.github.io

Systems engineer working on Cloudflare's Zero Trust data plane. Experience in network infrastructure, compiler instrumentation, security research, and performance analysis. Strong background in C/C++, Python, Go, and Linux systems.

EXPERIENCE

Cloudflare | System Engineer

Austin, TX · Nov 2025 - Present

- Contribute to the Zero Trust data plane, supporting policy-driven traffic routing across a global edge network of 50,000+ bare-metal servers handling billions of requests daily.
- Build and enhance networking components for TCP termination, regionalization, and security policy enforcement.
- Improve traffic-flow debugging through packet sampling and telemetry for routing decisions, performance, and policy enforcement.

InterSystems Corporation | Developer Support Engineer

Cambridge, MA · Jul 2023 - Nov 2025

- Diagnosed networking and kernel security issues in mission-critical InterSystems applications.
- Developed an ObjectScript performance-tracing API capturing CPU cycles, network connections, database accesses, and function calls.
- Modified the C++ ObjectScript compiler to insert instrumentation hooks during compilation.

Siemens | Software Development Intern

Princeton, NJ · Jun 2022 - Aug 2022

- Designed and prototyped a Zero Trust security extension for 5G core services using Free5GC, Go, and C++.
- Supported 100,000+ devices with less than 1% performance overhead in the prototype.

TECHNICAL SKILLS

Languages: C, C++, Python, Go, JavaScript

Systems & tools: Linux, Bash, Docker, eBPF, LLVM, x86, microservices, MySQL

Areas: TCP/IP, QUIC, Zero Trust, distributed systems, program analysis, fuzz testing

RESEARCH EXPERIENCE

Brown University | Research Assistant

Providence, RI · Jan 2021 - May 2023

- Built a Node.js syscall-filtering tool using dynamic program analysis, reducing exploit potential by 95% with 0-2% performance overhead.
- Developed distributed QUIC fuzz testing that discovered bugs in Meta and Cloudflare implementations.
- Compared QUIC and TCP performance: QUIC was up to 20% faster for smaller objects, with negligible differences for pages of 1 MB or larger.

Lahore University of Management Sciences | Research Assistant

Lahore, Pakistan · Jan 2019 - May 2020

- Investigated mobile video bottlenecks: memory pressure reduced frame rates by up to 80%; kernel memory management and disk I/O caused frame drops of up to 40%.

SELECTED PROJECTS

Static taint analysis. Implemented an LLVM pass in C++ to identify information leakage in Rust applications.

Kernel-level network monitoring. Built an eBPF tool for Ethernet, IP, and TCP packet inspection and service-level CPU utilization.

Protocol implementation. Implemented QUIC alongside UDP and Ethernet stack components in C.

EDUCATION

Brown University | Sc.M. in Computer Science

Providence, RI · May 2023 · GPA: 4.0/4.0

Coursework: distributed systems, compilers and program analysis, machine learning, applied cryptography, software security and exploitation.

Lahore University of Management Sciences | BS in Computer Science

Lahore, Pakistan · May 2020 · Graduated with Distinction

Focus: computer networking and security.

PUBLICATIONS

Extending 5G services with Zero Trust security pillars: a modular approach.
IEEE/ACS 19th International Conference on Computer Systems and Applications, 2022.

Mobile web browsing under memory pressure.
ACM SIGCOMM Computer Communication Review, 2020.